

Theorem 2.6.

Fix $K = (0, \mathbb{B})$ in $\mathbb{A}^m$.

The set S of points of $\mathbb{A}^m$ whose coord. are solutions to (s) is an affine space of dimension $d = m - r$.

$$(s) \begin{cases} a_{11}x_1 + \dots + a_{1m}x_m = b_1 \\ \vdots \\ a_{r1}x_1 + \dots + a_{rm}x_m = b_r. \end{cases}$$

$$D(s): \begin{cases} a_{11}x_1 + \dots + a_{1m}x_m = 0 \\ \vdots \\ a_{r1}x_1 + \dots + a_{rm}x_m = 0. \end{cases}$$

" $\Rightarrow$ " $S =$ set of solutions to (s)

$S = \{x_1, \dots, x_m\}$ (hyp.)

$S \rightarrow$ affine subspace of $\mathbb{A}^m$ (cl.)

$$\boxed{S = U + P.}$$

$P \rightarrow$ solution from S ; $P(p_1, \dots, p_m)$.

$U \rightarrow$ vector subspace.

$$\underline{S = T}$$

$$\hookrightarrow S \subseteq T.$$

$T \rightarrow$ sol. from S ;

$Q \rightarrow$ sol. from S ; $Q(q_1, \dots, q_m) \in S$.

$$\Rightarrow \boxed{Q \in T \Leftrightarrow \vec{PQ} \in U}$$

$$\begin{aligned} & a_{j1}(q_1 - p_1) + \dots + a_{jm}(q_m - p_m) = \\ & = \underbrace{a_{j1}q_1 + \dots + a_{jm}q_m}_{b_j} - \underbrace{(a_{j1}p_1 + \dots + a_{jm}p_m)}_{b_j} = 0 \end{aligned}$$

$$\Rightarrow \vec{PQ} \in U$$

$$\Rightarrow S \subseteq T$$

$$\begin{matrix} T \subseteq S \\ S \subseteq T \end{matrix} \Rightarrow S = T \Rightarrow S = U + P.$$

1) $\vec{BC}$ and $i \rightarrow$ orthogonal to each other?
 $\hookrightarrow$ DOT PRODUCT = SCALAR PRODUCT = 0 $\Rightarrow BC \cdot i = 0 \Rightarrow \langle \vec{BC}, \vec{i} \rangle = 0$

$$u \iff u$$

Define $f: \mathbb{R}^m \rightarrow \mathbb{W}$, $f(\alpha_1, \dots, \alpha_m) = \alpha_1 w_1 + \alpha_2 w_2 + \dots + \alpha_d w_d$ (lin. map.)

$$\boxed{\dim \mathbb{R}^m = \dim \text{Ker } f + \dim \text{Im } f}$$

$\text{Im } f = \langle w_1, \dots, w_d \rangle \rightarrow$ Gauss $\rightarrow$ echelon $\rightarrow$ ^{nr. of} $\neq 0$ lines $\Rightarrow \dim \text{Im } f$.

$$\text{Ker } f = \{(\alpha_1, \dots, \alpha_m) \in \mathbb{R}^m \mid f(\alpha_1, \dots, \alpha_m) = 0\}$$

$$= \{(\alpha_1, \dots, \alpha_m) \in \mathbb{R}^m \mid \langle \alpha_1 \underline{v}_1 + \dots + \alpha_m \underline{v}_m \rangle = 0\}$$

$$v_1 = \begin{pmatrix} v_{11} \\ v_{21} \\ \vdots \\ v_{m1} \end{pmatrix}, \dots, v_m = \begin{pmatrix} v_{1m} \\ v_{2m} \\ \vdots \\ v_{mm} \end{pmatrix}$$

$$\cancel{\dim \text{Ker } f} \Rightarrow \begin{cases} \alpha_1 \cdot \underline{v_{11}} + \dots + \alpha_m \cdot \underline{v_{1m}} = 0 \\ \vdots \\ \alpha_1 \cdot \underline{v_{m1}} + \dots + \alpha_m \cdot \underline{v_{mm}} = 0 \end{cases}$$

$$\underline{\dim \text{Ker } f = m - d} \Rightarrow \text{rank} \begin{pmatrix} v_{11} & \dots & v_{1m} \\ \vdots & & \vdots \\ v_{m1} & \dots & v_{mm} \end{pmatrix} = m - d.$$

$\Rightarrow$ Keep $m-d$ eq.

Theorem 2.6.

S affine subspace of A^m .

$$\dim S = d.$$

$\Rightarrow \exists$ system of $m-d$ linear equations in m unknowns whose solution

set is S .

$D(S) \Rightarrow$ vector subspace.

S affine subspace $\Rightarrow S = a + \langle v_1, v_2, \dots, v_m \rangle$,
of dim d .

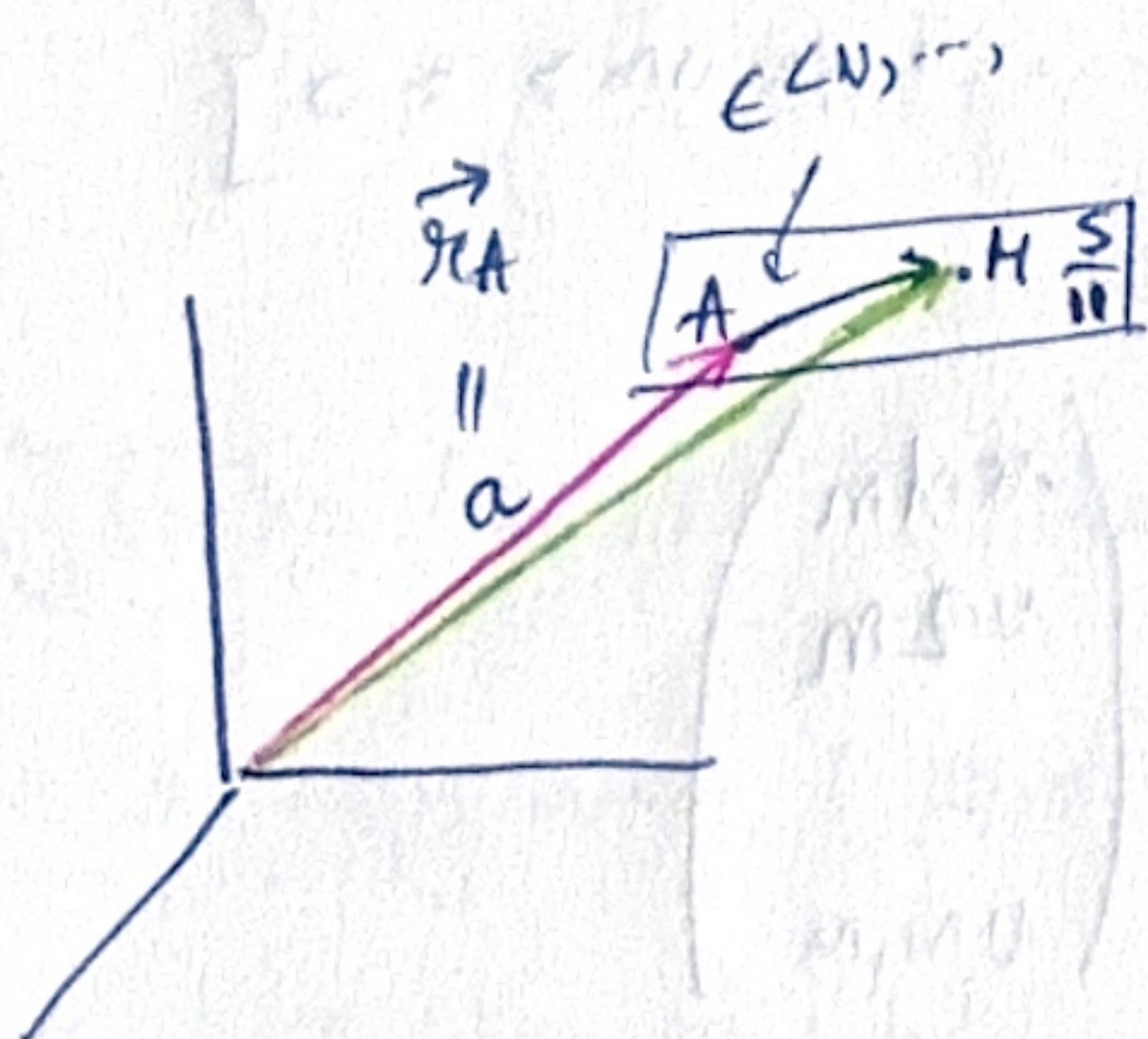
where $(v_i)_{i=1, \dots, d}$ lin. indep.

$$\Rightarrow d \leq m.$$

$a = (a_1, \dots, a_m) \rightarrow$ vector (point) $\rightarrow$ any point.

$$v_i = (v_{i1}, v_{i2}, \dots, v_{im}), (\forall) i.$$

$\leftarrow$ column.

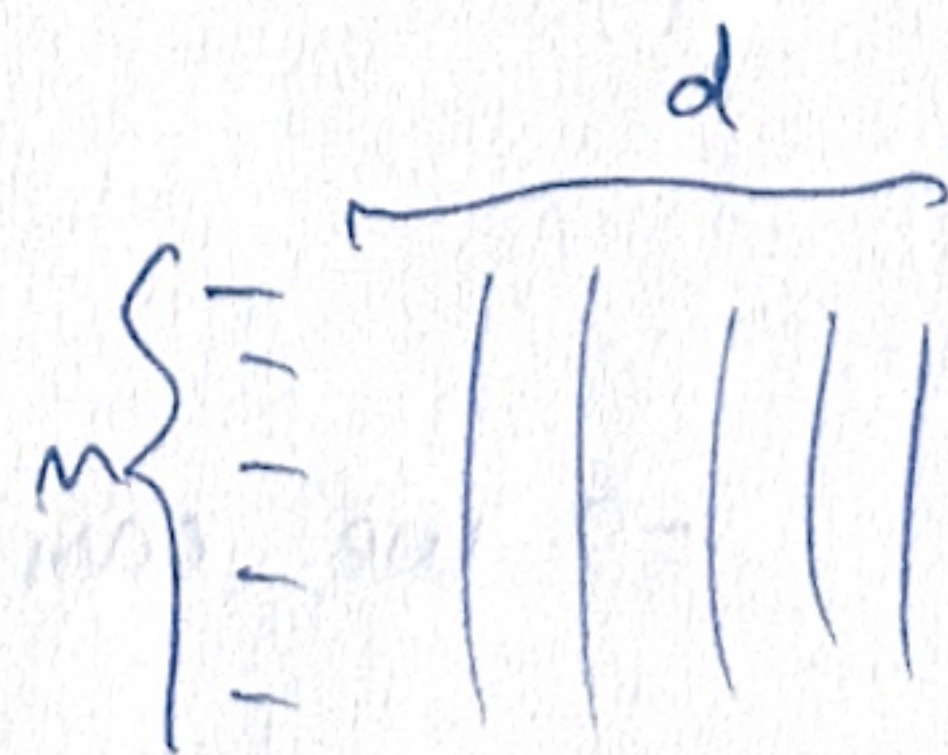


$$(x_1, \dots, x_m) \in S \Leftrightarrow \exists! \alpha_1, \alpha_2, \dots, \alpha_d \in \mathbb{R} \text{ s.t.}$$

$$\begin{cases} x_1 = a_1 + \alpha_1 \cdot v_{1,1} + \alpha_2 \cdot v_{1,2} + \dots + \alpha_d \cdot v_{1,d} \\ \vdots \\ x_m = a_m + \alpha_1 \cdot v_{m,1} + \alpha_2 \cdot v_{m,2} + \dots + \alpha_d \cdot v_{m,d} \end{cases}$$

unique sol. for each sol.

$$\text{rank} \begin{vmatrix} v_{1,1} & v_{1,2} & \dots & v_{1,d} \\ \vdots & \vdots & \vdots & \vdots \\ v_{m,1} & v_{m,2} & \dots & v_{m,d} \end{vmatrix} =$$



when has sol?

$$= \text{rank} \begin{vmatrix} v_{1,1} & v_{1,2} & \dots & v_{1,d} & x_1 - a_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ v_{m,1} & v_{m,2} & \dots & v_{m,d} & x_m - a_m \end{vmatrix} = d.$$

m rows

d cols.

$$d < m$$

If $W \leq V$, $\dim V = m$, $\dim W = d$, then W can be described by a system with

Define $f: \mathbb{R}^m \rightarrow V$

$$f(\alpha_1, \dots, \alpha_m) = \alpha_1 w_1 + \alpha_2 w_2 + \dots + \alpha_d w_d \quad (\text{linear map.})$$

$\rightarrow$ solve the system $\Rightarrow \dim$

$$\boxed{\dim \mathbb{R}^m = \dim \text{Ker } f + \dim \text{Im } f}$$

$$\text{Im } f = \langle w_1, \dots, w_d \rangle$$

$\rightarrow$ Gauss $\rightarrow$ echelon $\rightarrow \neq 0$ lines $\Rightarrow \dim \text{Im } f$

$$\dim \text{Ker } f = m - d.$$

lin. indep.

$$\text{Ker } f = \{ (\alpha_1, \dots, \alpha_m) \in \mathbb{R}^m \mid f(\alpha_1, \dots, \alpha_m) = 0 \}$$

$$= \{ (\alpha_1, \dots, \alpha_m) \in \mathbb{R}^m \mid \langle \alpha_1 w_1 + \dots + \alpha_m w_m \rangle = 0 \}$$

$$w_1 = \begin{pmatrix} w_{11} \\ w_{21} \\ \vdots \\ w_{m1} \end{pmatrix} \dots$$

$$w_m = \begin{pmatrix} w_{1m} \\ w_{2m} \\ \vdots \\ w_{mm} \end{pmatrix}$$

$$\Rightarrow \begin{cases} \alpha_1 w_{11} + \dots + \alpha_m w_{1m} = 0 \\ \vdots \\ \alpha_1 w_{m1} + \dots + \alpha_m w_{mm} = 0. \end{cases}$$

$$\dim \text{Ker } f = m - d \Rightarrow \text{rank} \begin{pmatrix} w_{11} & \dots & w_{1m} \\ \vdots & & \vdots \\ w_{m1} & \dots & w_{mm} \end{pmatrix} = m - d.$$

$\Rightarrow$ we can keep $m - d$ equations.

Related example:

$$f: \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad f(x, y, z) = (x + y, z)$$

1 0 0

$$\dim \text{Ker } f = ? \quad \dim \text{Im } f = ?$$

$$[f]_{EE'} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ already in echelon.}$$

$$\dim \text{Im } f = \text{rank } [f]_{EE'} = 2.$$

$$\dim \text{Ker } f = \{ (x, y, z) \in \mathbb{R}^3 \mid \begin{cases} x + y = 0 \\ z = 0 \end{cases} \Rightarrow \{ (x, x, 0), x \in \mathbb{R} \} \Rightarrow \dim \text{Ker } f = 1$$

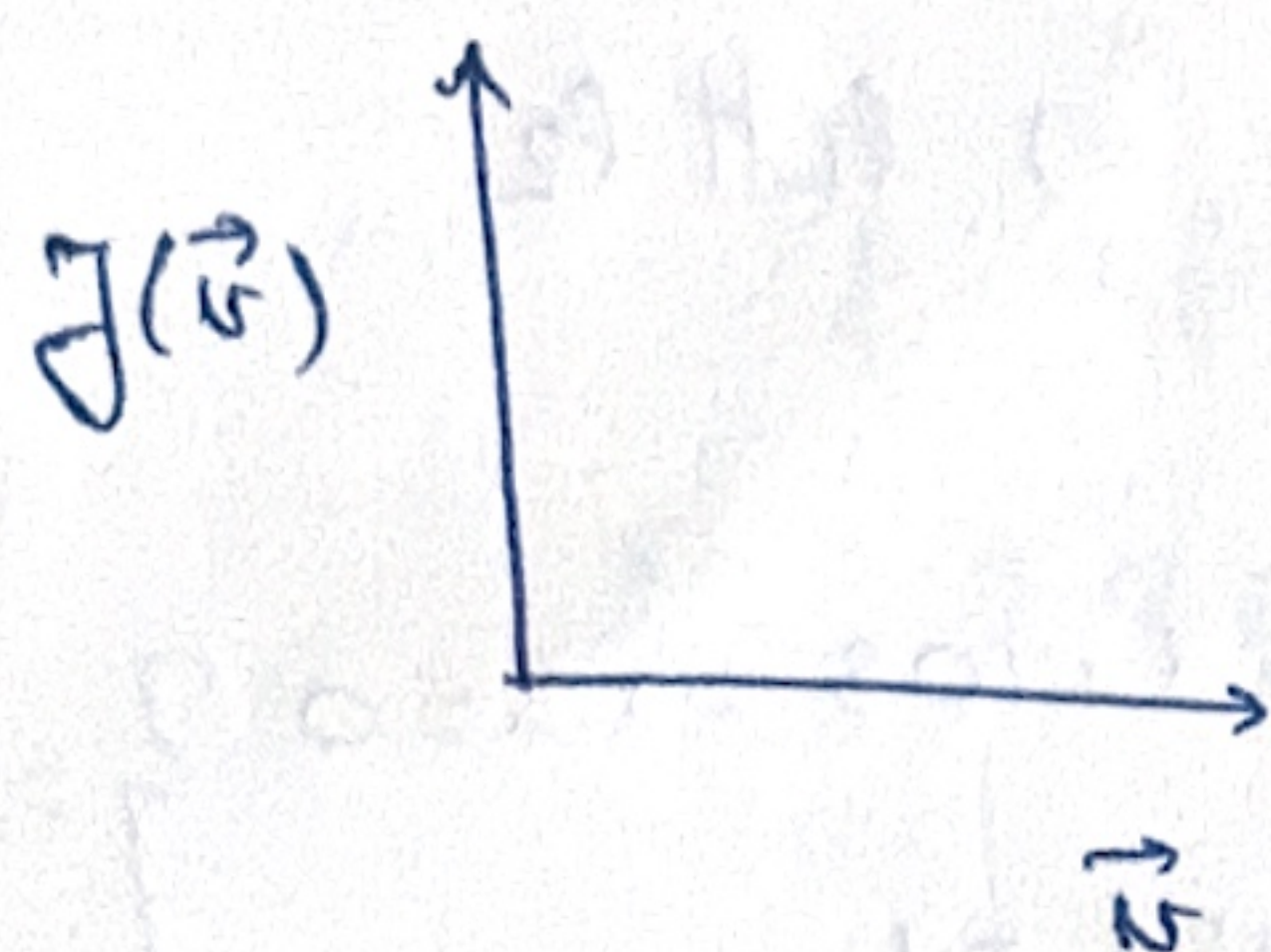
system has $m - d = 3 - 1$ eq.

the space

The pb: Prob 16

[3.8 Def.]

$\hat{u}(x, y)$ is an orthonormal basis.



rotations,
isometries

$$J(\hat{u}) = \begin{pmatrix} \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} \\ \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

$J(\hat{u}) = (a, b)$ for

$$\begin{aligned} ax + by &= 0 \quad (\text{due to } \perp) \Rightarrow b = -\frac{ax}{y}, y \neq 0 \\ x^2 + y^2 &= a^2 + b^2 \end{aligned} \Rightarrow \begin{aligned} x^2 + y^2 &= a^2 + \frac{a^2 \cdot x^2}{y^2} \\ \Rightarrow x^2 + y^2 &= a^2 \left(1 + \frac{x^2}{y^2}\right) \end{aligned}$$

$(u, J(u))$ right oriented. $(\Rightarrow)$ $\begin{vmatrix} x & a \\ y & b \end{vmatrix} > 0$.

$$\Rightarrow x^2 + y^2 = a^2 - \frac{x^2 + y^2}{b^2}$$

$$x^2 + y^2 \neq 0 \text{ bc } \hat{u} \neq \hat{0}$$

$$\Rightarrow a^2 = y^2$$

$$\Rightarrow a = \pm y.$$

$$\Rightarrow \begin{vmatrix} x & a \\ y & b \end{vmatrix} = \begin{vmatrix} x & \pm y \\ y & \mp y - \frac{x}{y} \end{vmatrix} = \begin{vmatrix} x & \pm y \\ y & \mp x \end{vmatrix}$$

$$= \mp (x^2 + y^2) > 0$$

$$\Rightarrow a = -$$

! [Def: 1.15] Left / Right oriented.

smooth curve in $\mathbb{R}^2$
(counterclockwise)

Quiz 2, Ex. 5. Relative pos. of:

lim. dep $\Rightarrow \parallel$

Sol: $D(l_1) = \langle (-3, 3, 0) \rangle$
 $D(l_2) = \langle (5, 13, 0) \rangle$ $\Rightarrow$ not parallel.

$D(l_1) \not\subseteq D(l_2)$ and $D(l_2) \subseteq D(l_1) \Rightarrow l_1 \nparallel l_2$.

$l_1 \cap l_2$:

$$\begin{cases} x = -3t \\ y = 2 + 3t \\ z = 1 \end{cases}$$

$$\begin{cases} x = 1 + 5s \\ y = 1 + 13s \\ z = 1 + 10s \end{cases}$$

(2) $z = 1 = 1 + 10s \Rightarrow s = 0$
 $x = -3t = 1 \Rightarrow t = -\frac{1}{3}$
 $y = 2 + 3t = 1 + 13s$
 $\Rightarrow y = 2 + 3 \cdot (-\frac{1}{3}) = 1 + 13 \cdot 0 = 1$

$\Rightarrow l_1 \cap l_2 = \{(1, 1, 1)\}$. ID
 (incident in one point)

⑥ A, B, C, D coplanar $\Leftrightarrow$

$$\begin{vmatrix} x_A & y_A & z_A & 1 \\ x_B & y_B & z_B & 1 \\ x_C & y_C & z_C & 1 \\ x_D & y_D & z_D & 1 \end{vmatrix} = 0.$$

(2) $D \in (ABC)$

(ABC):

$$\begin{vmatrix} x & y & z & 1 \\ x_A & y_A & z_A & 1 \\ x_B & y_B & z_B & 1 \\ x_C & y_C & z_C & 1 \end{vmatrix} = 0 \Leftrightarrow (ABC) \neq \emptyset$$

(2)

$$\begin{vmatrix} x - x_A & y - y_A & z - z_A \\ x_B - x_A & y_B - y_A & z_B - z_A \\ x_C - x_A & y_C - y_A & z_C - z_A \end{vmatrix} = 0.$$

$$\begin{vmatrix} x - x_A & y - y_A & z - z_A \\ x_B - x_A & y_B - y_A & z_B - z_A \\ x_C - x_A & y_C - y_A & z_C - z_A \end{vmatrix} = 0.$$

3 points $\Rightarrow$ plane
 (horse statues)

$$\begin{vmatrix} x - x_A & y - y_A & z - z_A & 0 \\ x_B - x_A & y_B - y_A & z_B - z_A & 0 \\ x_C - x_A & y_C - y_A & z_C - z_A & 0 \end{vmatrix} =$$

(15) Sol: If the ^{affine} subspace is a point Q , then:

$$\begin{aligned} \mathcal{L} &: \{P \in \mathbb{E}^2 \mid \text{dist}(P, Q) = \alpha\} = \\ &= \{P(x, y) \mid \sqrt{(x-x_Q)^2 + (y-y_Q)^2} = \alpha\} = \text{circle}. \end{aligned}$$

If affine subsp. is a line l , then:
 $l: ax + by + c = 0$

$$\begin{aligned} \mathcal{L} &: \{P \in \mathbb{E}^2 \mid \text{dist}(P, l) = \alpha\} \\ &= \left\{P(x, y) \mid \frac{|ax + by + c|}{\sqrt{a^2 + b^2}} = \alpha\right\} \\ &= \{P \mid ax + by + c = \pm \alpha \sqrt{a^2 + b^2}\} \\ &= \{P \mid ax + by + c + \alpha \sqrt{a^2 + b^2} = 0\} \cup \\ &\quad \{P \mid ax + by + c - \alpha \sqrt{a^2 + b^2} = 0\} \\ &= 2 \text{ lines}. \end{aligned}$$